

LIOUVILLE AND LYAPUNOV CRITERIA FOR ANCIENT NAVIER–STOKES SOLUTIONS

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ABSTRACT. We study bounded ancient solutions of the three-dimensional Navier–Stokes equations arising from Type I blow-up arguments. We state standard PDE hypotheses under which such solutions are spatially constant; in the Type I class, these constants vanish. We also prove two Lyapunov rigidity results for rescaled vorticity equations.

For the unforced equations, known axisymmetric Liouville theorems cover the following cases: absence of swirl, pointwise scale-invariant control, finite $L_t^\infty L_x^p$ control of the swirl variable $\Gamma = ru_\theta$, and axial periodicity with bounded Γ . When one of these hypotheses holds for a nonzero Type I blow-up limit, the Galilean-trivial conclusions of the cited Liouville theorems reduce to the zero solution and contradict the nontriviality of the limit.

We also record two Lyapunov rigidity arguments. First, Gallay–Wayne’s small-data decay theorem for the three-dimensional rescaled vorticity equation yields a coercive Lyapunov functional forcing any ancient trajectory that stays in the perturbative weighted-vorticity ball to be zero. Second, Gallay–Maekawa’s stability theorem for the strained Burgers-vortex equation yields a Lyapunov functional for perturbations of a Burgers vortex and rigidity of ancient solutions in a small weighted neighborhood of the Burgers-vortex family.

Finally, for nonperiodic bounded-swirl axisymmetric solutions, we prove the exact scalar Γ -identity and the anisotropic cutoff-error decomposition. These identities give a coercive expanding-cutoff criterion that forces Γ to vanish, reducing the solution to the no-swirl case. The paper concludes with a summary of the Liouville hypotheses, conditional criteria, and cited PDE results.

1. INTRODUCTION

1.1. Ancient solutions from Type I blow-up. The Type I blow-up argument in Paper I [3] produces, after parabolic rescaling around a Type I singular point, a nonzero ancient solution (U, P) of

$$(1.1) \quad \partial_t U + (U \cdot \nabla)U + \nabla P = \Delta U, \quad \nabla \cdot U = 0, \quad t < 0,$$

satisfying the Type I ancient bound

$$(1.2) \quad \sup_{t < 0} \sqrt{-t} \|U(t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Thus every result proving $U \equiv 0$ for an ancient solution satisfying (1.2) has the same consequence in the Type I problem:

$$\text{PDE hypothesis} \implies U \equiv 0 \implies \text{contradiction with nonzero blow-up limit.}$$

The paper treats hypotheses for which a specific Liouville mechanism is available, or for which a precise conditional criterion can be stated. The cases considered are:

axisymmetric Liouville hypotheses, small weighted rescaled vorticity,
the strained Burgers-vortex equation, bounded swirl with expanding cutoff control.

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We make no assertion for decaying or symmetry-restricted ancient solutions outside the hypotheses listed below. Each conclusion is tied to an explicit PDE hypothesis and a specified analytic result.

For the Type I application in Paper I, only those cases in which the stated PDE hypothesis and the cited or proved Liouville theorem imply $U \equiv 0$ are used as exclusions. Ancient profiles outside these hypotheses are treated separately.

1.2. Axisymmetric Liouville hypotheses. Several unforced axisymmetric cases are already covered by known Liouville theorems. Koch–Nadirashvili–Seregin–Šverák cover the no-swirl case and the pointwise scale-invariant condition $|u| \leq C/r$. The swirl-bearing finite- L^p condition

$$\Gamma = ru_\theta \in L_t^\infty L_x^p, \quad 1 \leq p < \infty,$$

is covered by Lei–Zhang–Zhao. The z -periodic case with bounded Γ is covered by Lei–Ren–Zhang.

The only additional step needed for Type I ancient limits is to convert Galilean-triviality into zero in the blow-up frame. We time-shift away from the possible singular time $t = 0$, apply the bounded ancient Liouville theorem, patch the resulting constants on backward intervals, and use (1.2): if $U(x, t) \equiv c$, then

$$\sup_{t < 0} \sqrt{-t} \|U(t)\|_\infty = \sup_{t < 0} \sqrt{-t} |c|$$

is finite only when $c = 0$.

1.3. Lyapunov hypotheses. The two Lyapunov results below use forward Lyapunov functionals. The first concerns the Gallay–Wayne perturbative weighted-vorticity regime. If w solves the rescaled three-dimensional vorticity equation and remains in the Gallay–Wayne small ball of $L_\sigma^2(m)$, $m > 7/2$, then the functional

$$(1.3) \quad \mathcal{L}_m(w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2$$

is coercive and contracts along the forward semiflow. Sending the starting time to $-\infty$ then forces every ancient trajectory in the small ball to be zero.

The second Lyapunov argument is for the strained Burgers-vortex equation

$$\partial_\tau \Omega + (U \cdot \nabla) \Omega - (\Omega \cdot \nabla) U = \Delta \Omega - (Ax \cdot \nabla) \Omega + A\Omega, \quad A = \text{diag} \left(-\frac{1}{2}, -\frac{1}{2}, 1 \right),$$

not directly for the unforced Navier–Stokes equations. Gallay–Maekawa’s stability theorem gives a Lyapunov functional for perturbations with fixed circulation and forces ancient trajectories in a small weighted neighborhood of the Burgers-vortex family to be exactly stationary Burgers vortices. Applying that conclusion to an unforced Navier–Stokes solution requires the additional association assumption stated later.

1.4. The bounded-swirl case. The remaining axisymmetric case treated here is bounded swirl without finite- L^p decay or axial periodicity:

$$\Gamma = ru_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

For a smooth axisymmetric solution, the scalar Γ satisfies

$$(1.4) \quad \partial_t \Gamma + b \cdot \nabla \Gamma + \frac{2}{r} \partial_r \Gamma = \Delta \Gamma, \quad b = u_r e_r + u_z e_z.$$

The corresponding weighted identity is

$$(1.5) \quad \frac{1}{2} \frac{d}{dt} \int \Gamma^2 \phi \, dx + \int |\nabla \Gamma|^2 \phi \, dx = \frac{1}{2} \int \Gamma^2 \left(\partial_t \phi + \Delta \phi + b \cdot \nabla \phi + \frac{2}{r} \partial_r \phi \right) \, dx.$$

For anisotropic product cutoffs

$$\phi_{R,Z,\zeta}(r, z, t) = \lambda_R(r) \eta_Z(z - \zeta(t)),$$

the error density splits exactly into radial cutoff transport, axial drift mismatch, and cutoff curvature. The radial diffusion operator is $\partial_r^2 + 3r^{-1} \partial_r$, the radial Laplacian in four dimensions. We prove that a coercive expanding-cutoff hypothesis forces $\Gamma \equiv 0$. No assertion is made that every bounded nonperiodic Γ satisfies this additional hypothesis.

1.5. Main results. The first result records the unforced axisymmetric Liouville consequences relevant to the Type I argument.

Theorem 1.1 (Axisymmetric Liouville consequences for Type I ancient solutions). *Let U be a smooth Type I ancient solution of (1.1) satisfying (1.2). If U satisfies any of the following axisymmetric assumptions, then $U \equiv 0$:*

- (i) *axisymmetric without swirl;*
- (ii) *axisymmetric and satisfying a pointwise scale-invariant bound*

$$|U(x, t)| \leq \frac{C}{r};$$

- (iii) *axisymmetric with*

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty;$$

- (iv) *axisymmetric, periodic in the physical axial variable z , and with*

$$\Gamma = rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Consequently none of these assumptions can hold for a nonzero Type I blow-up limit.

Theorem 1.2 (Perturbative weighted-vorticity Lyapunov rigidity). *Let $m > \frac{7}{2}$. Let $w(\tau)$ be an ancient mild solution of the Galloway–Wayne rescaled vorticity equation*

$$(1.6) \quad \partial_\tau w = \Lambda w - (v \cdot \nabla)w + (w \cdot \nabla)v, \quad \nabla \cdot w = 0,$$

with v recovered by the three-dimensional Biot–Savart law. Assume

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_{L_o^2(m)} \leq \rho_m,$$

where ρ_m is the Galloway–Wayne small-data radius. Then

$$w \equiv 0.$$

Here “ancient mild” means that every forward time-shift of w is the Galloway–Wayne mild solution launched from its value at the starting time.

Theorem 1.3 (Ancient rigidity near Burgers vortices). *Let $m > 2$. Let $\Omega : \mathbb{R} \rightarrow \mathcal{X}_{\text{full}}(m)$ be an admissible ancient mild solution of the strained Burgers-vortex vorticity equation*

$$(1.7) \quad \partial_\tau \Omega + (U \cdot \nabla) \Omega - (\Omega \cdot \nabla) U = \Delta \Omega - (A x \cdot \nabla) \Omega + A \Omega, \quad A = \text{diag} \left(-\frac{1}{2}, -\frac{1}{2}, 1 \right).$$

Assume Ω remains in the Galloway–Maekawa small weighted neighborhood of the Burgers-vortex family, where $\bar{\alpha}$ denotes the circulation of Ω :

$$\sup_{\tau \in \mathbb{R}} \|\Omega(\tau) - \bar{\alpha} G\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\bar{\alpha}, m}.$$

Then

$$\Omega(\tau) \equiv \bar{\alpha}G \quad \text{for all } \tau \in \mathbb{R}.$$

This theorem concerns the strained Burgers-vortex equation. It gives a statement about unforced Navier–Stokes ancient limits only under the additional association assumption stated in [Theorem 7.7](#). The term “ancient mild” means compatibility with the forward Gally–Maekawa semiflow after every time shift, as made explicit in [Theorem 7.6](#).

Theorem 1.4 (Localized scalar criterion for Γ). *Let u be a smooth bounded ancient mild axisymmetric solution of the unforced Navier–Stokes equations with bounded swirl variable*

$$\Gamma = ru_\theta \in L^\infty.$$

Assume the coercive expanding-cutoff hypotheses stated in [Theorem 8.5](#). Then

$$\Gamma \equiv 0 \quad \text{on } \mathbb{R}^3 \times (-\infty, 0).$$

Consequently, if the resulting no-swirl solution is covered by the standard axisymmetric no-swirl Liouville theorem, then u is Galilean-trivial. If, in addition, u is a Type I blow-up limit, then $u \equiv 0$.

Theorem 1.5 (Consequences for Type I blow-up limits). *Let U be a nonzero Type I blow-up limit. Then U cannot satisfy any of the axisymmetric assumptions in [Theorem 1.1](#); nor can its corresponding rescaled vorticity satisfy the hypotheses of [Theorem 1.2](#). The solution also cannot satisfy the bounded-swirl assumptions together with [Theorem 8.5](#). Separately, if a field Ω_U satisfies the assumptions in [Theorem 7.7](#), then $\Omega_U(\tau) \equiv \bar{\alpha}G$. This last conclusion is a rigidity statement for the strained Burgers-vortex equation and does not imply $U \equiv 0$ for the unforced Navier–Stokes equations.*

Only the explicit hypotheses stated above are used.

1.6. Relation to Papers I and II. Paper I [\[3\]](#) reduces Type I singularities to nonzero ancient limits satisfying [\(1.2\)](#). Paper II [\[4\]](#) treats uniformly L^3 -tight ancient solutions. The present paper treats the symmetry, decay, and Burgers-vortex hypotheses listed below. Known Liouville theorems rule out several cases, Lyapunov arguments handle perturbative cases, and the remaining bounded-swirl axisymmetric case is reduced to a precise expanding-cutoff criterion. Paper III does not depend on Paper II.

1.7. Summary of hypotheses.

Hypothesis	Conclusion	Analytic ingredient
axisymmetric no swirl	$U \equiv 0$	KNSS
pointwise $ u \leq C/r$	$U \equiv 0$	KNSS
$\Gamma \in L_t^\infty L_x^p$, $1 \leq p < \infty$	$U \equiv 0$	LZZ
z -periodic bounded Γ	$U \equiv 0$	LRZ
controlled swirl $\rho^\gamma V_\theta$	conditional	fixed-exponent Liouville hypothesis
weighted velocity decay	conditional	fixed-exponent Liouville hypothesis
small weighted vorticity	$w \equiv 0$	Gally–Wayne
near Burgers vortices	$\Omega = \alpha G$	Gally–Maekawa
association with Burgers vortices	conditional	Theorem 7.7
bounded nonperiodic $\Gamma \in L^\infty$	conditional	cutoff criterion needed
bounded Γ plus cutoff control	$\Gamma \equiv 0$	scalar identity plus no-swirl theorem

1.8. Organization. Section 2 records the Type I ancient setup, self-similar variables, the invariance facts, and the time-shift lemma. Section 3 fixes axisymmetric notation and the known Liouville theorems. Section 4 proves the unforced axisymmetric Liouville consequences. Section 5 records the controlled swirl and weighted velocity-decay conditions as conditional hypotheses. Section 6 proves the Gallay–Wayne Lyapunov rigidity. Section 7 proves rigidity for the Burgers-vortex equation. Section 8 derives the scalar Γ -identity, cutoff-error decomposition, and localized scalar theorem. Section 9 summarizes the consequences.

2. ANCIENT TYPE I LIMITS AND SELF-SIMILAR VARIABLES

2.1. Type I ancient solutions.

Definition 2.1 (Type I ancient solution). A smooth mild solution U of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$ is called Type I ancient if (1.2) holds. Mild is understood in the standard velocity formulation [9, 17]: for every $-\infty < s < t < 0$,

$$(2.1) \quad U(t) = e^{(t-s)\Delta}U(s) - \int_s^t e^{(t-\sigma)\Delta}\mathbb{P}\nabla \cdot (U(\sigma) \otimes U(\sigma)) d\sigma$$

in distributions, where $\mathbb{P}$ is the Leray projector.

2.2. Nonzero Type I blow-up limits.

Definition 2.2 (Nonzero Type I blow-up limit). A Type I ancient solution U is called a nonzero Type I blow-up limit if it arises as the limit of parabolic rescalings around a Type I singular point and the limiting solution is not identically zero. The arguments below require only the conclusion $U \not\equiv 0$.

Lemma 2.3 (Nontriviality of blow-up limits). *A nonzero Type I blow-up limit is not identically zero.*

Proof. This is part of the definition above. □

2.3. Backward self-similar variables.

For $t < 0$, define

$$(2.2) \quad V(y, \tau) = \sqrt{-t}U(x, t), \quad Q(y, \tau) = (-t)P(x, t), \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t).$$

Equivalently,

$$x = e^{-\tau/2}y, \quad t = -e^{-\tau}.$$

Lemma 2.4 (Self-similar equation). *Let (U, P) be a smooth solution of (1.1), and define (V, Q) by (2.2). Then V is defined on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies*

$$(2.3) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla Q = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \nabla \cdot V = 0.$$

Moreover (1.2) is equivalent to

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Proof. Let $a = \sqrt{-t} = e^{-\tau/2}$. Then $V = aU$, $y = x/a$, and

$$\partial_t \tau = \frac{1}{-t} = \frac{1}{a^2}, \quad \partial_t y = \frac{y}{2a^2}, \quad \partial_t(a^{-1}) = \frac{1}{2a^3}.$$

Since $U = a^{-1}V$, differentiating at fixed x gives

$$\partial_t U = a^{-3} \left(\partial_\tau V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V \right).$$

Spatial derivatives scale as

$$\nabla_x U = a^{-2} \nabla_y V, \quad \Delta_x U = a^{-3} \Delta_y V,$$

and the pressure scaling gives

$$\nabla_x P = a^{-3} \nabla_y Q.$$

Thus

$$(U \cdot \nabla_x) U = a^{-3} (V \cdot \nabla_y) V.$$

Multiplying (1.1) by a^3 yields (2.3); the divergence equation follows from $\nabla_x \cdot U = a^{-2} \nabla_y \cdot V$. The spatial map $y \mapsto x = e^{-\tau/2} y$ is a bijection for each τ , and

$$\|V(\tau)\|_\infty = \sqrt{-t} \|U(t)\|_\infty.$$

Taking suprema over $t < 0$, equivalently $\tau \in \mathbb{R}$, gives the last assertion. $\square$

2.4. Preservation of axisymmetry and swirl variables. For axisymmetric flows we write

$$x = (r \cos \theta, r \sin \theta, z), \quad U = U_r e_r + U_\theta e_\theta + U_z e_z,$$

and in centered variables

$$y = (\rho \cos \theta, \rho \sin \theta, Z), \quad V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The scalar swirl variables are

$$\Gamma_U(r, z, t) = r U_\theta(r, z, t), \quad \Gamma_V(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau).$$

Lemma 2.5 (Preservation of axisymmetric structure). *Axisymmetry and absence of swirl are preserved under (2.2). If U is axisymmetric, then*

$$\Gamma_V(\rho, Z, \tau) = \Gamma_U(r, z, t), \quad (r, z) = \sqrt{-t}(\rho, Z).$$

Fixed physical z -periodicity is preserved by physical time shifts, but under the centered change of variables it becomes a τ -dependent period in the Z -coordinate.

Proof. The self-similar map is a scalar spatial dilation and a scalar multiplication of the velocity, so it commutes with rotations around the symmetry axis. The cylindrical basis is unchanged by radial dilation, and

$$V_\theta(\rho, Z, \tau) = \sqrt{-t} U_\theta(r, z, t), \quad r = \sqrt{-t} \rho, \quad z = \sqrt{-t} Z.$$

Thus $U_\theta \equiv 0$ if and only if $V_\theta \equiv 0$, and

$$\Gamma_V = \rho V_\theta = \rho \sqrt{-t} U_\theta = r U_\theta = \Gamma_U.$$

If U has physical axial period L , then V has period $L/\sqrt{-t} = L e^{\tau/2}$ in Z , not a fixed period. $\square$

2.5. Time shifts away from the singular time.

Lemma 2.6 (Time shifts away from the singular time). *Let U be a smooth Type I ancient solution and fix $T < 0$. Then*

$$W_T(x, s) = U(x, T + s), \quad \Pi_T(x, s) = P(x, T + s), \quad s < 0,$$

is a smooth bounded mild ancient solution of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$. Moreover the axisymmetric, no-swirl, L^p -swirl, bounded- Γ , and physical z -periodic properties are preserved.

Proof. Time translation preserves the differential equation because $\partial_s W_T(x, s) = \partial_t U(x, T + s)$ and all spatial derivatives are unchanged. For the mild formulation, apply (2.1) to U between the physical times $T + s < T + t < 0$:

$$U(T + t) = e^{(t-s)\Delta} U(T + s) - \int_{T+s}^{T+t} e^{(T+t-\eta)\Delta} \mathbb{P} \nabla \cdot (U(\eta) \otimes U(\eta)) d\eta.$$

With $\eta = T + \sigma$, this becomes the mild formula for W_T on the interval (s, t) . For $s < 0$, $T + s < T < 0$, so the Type I bound gives

$$\|W_T(s)\|_\infty = \|U(T + s)\|_\infty \leq \frac{1}{\sqrt{-T}} \sup_{\eta < 0} \sqrt{-\eta} \|U(\eta)\|_\infty.$$

The invariance assertions follow because only the time variable has been translated. In particular

$$\Gamma_{W_T}(r, z, s) = \Gamma_U(r, z, T + s),$$

so $L_s^\infty L_x^p$ and L^∞ bounds for Γ restrict to the shifted solution, and a fixed physical axial period remains the same. $\square$

2.6. PDE hypotheses used below. The arguments below are stated directly in terms of hypotheses on ancient solutions. The unconditional axisymmetric hypotheses are:

- (i) axisymmetry and $U_\theta \equiv 0$;
- (ii) axisymmetry and the pointwise bound $|U(x, t)| \leq C/r$ for $r > 0$;
- (iii) axisymmetry and

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty;$$

- (iv) axisymmetry, periodicity in the physical axial variable z , and $\Gamma \in L^\infty$.

The additional hypotheses used later are the fixed-exponent controlled-swirl Liouville statement, the fixed-exponent weighted-velocity Liouville statement, the Gallay–Wayne small weighted-vorticity condition, the association with the Burgers-vortex equation, and the coercive expanding-cutoff condition [Theorem 8.5](#).

3. AXISYMMETRIC NOTATION AND KNOWN LIOUVILLE THEOREMS

3.1. Cylindrical variables and the swirl scalar. For an axisymmetric solution u , write

$$u = u_r e_r + u_\theta e_\theta + u_z e_z, \quad \Gamma = ru_\theta, \quad b = u_r e_r + u_z e_z.$$

When Γ is placed in $L^p(\mathbb{R}^3)$, it is regarded as the corresponding function of $x = (r \cos \theta, r \sin \theta, z)$, independent of θ :

$$\|\Gamma(t)\|_{L^p(\mathbb{R}^3)}^p = 2\pi \int_{\mathbb{R}} \int_0^\infty |\Gamma(r, z, t)|^p r dr dz, \quad 1 \leq p < \infty.$$

3.2. Constant and Galilean-trivial solutions.

Definition 3.1 (Galilean-trivial). A velocity field is Galilean-trivial if, after a Galilean transform, it is a constant vector field. In the Type I blow-up frame below we use only the following consequence: when a cited Liouville theorem gives a constant velocity field in that frame, the Type I bound forces the constant to be zero.

Lemma 3.2 (Spatially constant mild solutions). *Let W be a smooth mild solution of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$. If $W(x, t) = b(t)$ for all x and t , then b is constant in time.*

Proof. Insert $W(x, t) = b(t)$ into (2.1). The heat semigroup leaves constant vector fields fixed, and $\nabla \cdot (b(\sigma) \otimes b(\sigma)) = 0$. Thus $b(t) = b(s)$ for every $s < t < 0$. $\square$

Lemma 3.3 (Patching constants on backward intervals). *Let $U : \mathbb{R}^3 \times (-\infty, 0) \rightarrow \mathbb{R}^3$ be a function. Assume that for every $T < 0$ there exists $c_T \in \mathbb{R}^3$ such that*

$$U(x, t) = c_T \quad (x \in \mathbb{R}^3, t < T).$$

Then there is a single $c \in \mathbb{R}^3$ such that

$$U(x, t) = c \quad (x \in \mathbb{R}^3, t < 0).$$

Proof. If $T_1 < T_2 < 0$, the two constants agree on the nonempty interval $(-\infty, T_1)$. Hence all c_T are equal. For any $t < 0$, choose T with $t < T < 0$. $\square$

Lemma 3.4 (Constant Type I ancient solutions vanish). *Let U be a Type I ancient solution. If $U(x, t) \equiv c$, then $c = 0$.*

Proof. The Type I bound gives

$$\sqrt{-t} |c| \leq \sup_{s < 0} \sqrt{-s} \|U(s)\|_\infty < \infty \quad (t < 0).$$

Letting $t \rightarrow -\infty$ gives $c = 0$. $\square$

3.3. Known Liouville theorems. We use the following Liouville theorems in the stated forms.

Theorem 3.5 (KNSS no-swirl Liouville theorem). *Every bounded ancient mild axisymmetric no-swirl solution of the unforced Navier–Stokes equations is Galilean-trivial. In the form used below, the solution is a spatially constant axial drift.*

Proof. This is the no-swirl Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [10], stated in the form needed for the Type I vanishing argument. $\square$

Theorem 3.6 (KNSS pointwise scale-invariant condition). *Every bounded ancient mild axisymmetric solution satisfying*

$$|u(x, t)| \leq \frac{C}{r} \quad (r > 0, t < 0)$$

is Galilean-trivial. In the form used below, it is identically zero.

Proof. This is the pointwise Liouville theorem of [10], stated in the form used here. $\square$

Theorem 3.7 (Lei–Zhang–Zhao). *Let u be a bounded ancient mild axisymmetric solution. If*

$$\Gamma = ru_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty,$$

then u is a constant vector field.

Proof. This is the ancient axisymmetric L^p -swirl Liouville theorem of Lei–Zhang–Zhao [13]. $\square$

Theorem 3.8 (Lei–Ren–Zhang). *Let u be a bounded ancient mild axisymmetric solution, periodic in z , with*

$$\Gamma = ru_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Then u is a constant axial vector field.

Proof. This is the periodic bounded-swirl theorem of Lei–Ren–Zhang [12]. Their statement is on $\mathbb{R}^2 \times \mathbb{T}^1$; a physical z -periodic whole-space solution descends to the quotient, and a Navier–Stokes scaling normalizes the period. $\square$

4. UNFORCED AXISYMMETRIC LIOUVILLE CONSEQUENCES

4.1. No swirl.

Proposition 4.1 (No-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and $U_\theta \equiv 0$. Then $U \equiv 0$.*

Proof. Fix $T < 0$ and set $W_T(x, s) = U(x, T + s)$. By Theorem 2.6, W_T is a bounded ancient mild axisymmetric no-swirl solution. Applying Theorem 3.5 gives that W_T is a spatially constant axial drift, say $W_T(x, s) = b_T(s)e_z$. By Theorem 3.2, $b_T(s)$ is constant in s . Thus $U(x, t) = c_T$ for $t < T$. Since $T < 0$ is arbitrary, Theorem 3.3 gives $U(x, t) = c$ for all $t < 0$. Finally, Theorem 3.4 gives $c = 0$. $\square$

4.2. Pointwise scale-invariant control.

Proposition 4.2 (Pointwise $1/r$ Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and satisfies*

$$|U(x, t)| \leq \frac{C}{r} \quad (r > 0, t < 0).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. The shifted solution $W_T(x, s) = U(x, T + s)$ is bounded, ancient, axisymmetric, and satisfies the same pointwise bound $|W_T| \leq C/r$. Theorem 3.6 gives $W_T \equiv 0$. Hence $U \equiv 0$ on $(-\infty, T)$. Since $T < 0$ is arbitrary, $U \equiv 0$. $\square$

 4.3. Finite L^p control of Γ .

Proposition 4.3 (Finite- L^p swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and, for some $1 \leq p < \infty$,*

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Theorem 2.6, $W_T(x, s) = U(x, T + s)$ is a bounded ancient mild axisymmetric solution and

$$\Gamma_{W_T} \in L_s^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

Theorem 3.7 gives that W_T is a constant vector field. Patch these constants in T by Theorem 3.3 and then use Theorem 3.4. $\square$

4.4. Periodic bounded swirl.

Proposition 4.4 (Periodic bounded-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric, periodic in the physical axial variable z , and satisfies*

$$\Gamma = rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Theorem 2.6, the shifted solution W_T is bounded, ancient, axisymmetric, physically z -periodic, and has bounded Γ . Theorem 3.8 gives that W_T is a constant axial vector field. The constants patch as T varies, and the Type I bound eliminates the resulting constant by Theorem 3.4. $\square$

4.5. Proof of the axisymmetric Liouville theorem.

Proof of Theorem 1.1. The four cases are precisely the hypotheses of Theorems 4.1 to 4.4. Each proposition gives $U \equiv 0$. This contradicts Theorem 2.3 when U is a nonzero Type I blow-up limit. $\square$

5. CONDITIONAL CONTROLLED-SWIRL AND WEIGHTED-DECAY HYPOTHESES

5.1. Controlled swirl in self-similar variables.

Definition 5.1 (Controlled-swirl condition). Fix $\gamma > 0$. A centered ancient solution V satisfies the controlled-swirl condition with exponent γ if V is axisymmetric and

$$\sup_{\tau \in \mathbb{R}} \sup_{\rho > 0, Z \in \mathbb{R}} \rho^\gamma |V_\theta(\rho, Z, \tau)| < \infty.$$

Hypothesis 5.2 (Fixed-exponent controlled-swirl Liouville theorem). For the chosen exponent $\gamma > 0$, every Type I ancient solution whose centered representative satisfies the controlled-swirl condition with exponent γ is identically zero.

Proposition 5.3 (Conditional controlled-swirl vanishing). *If Theorem 5.2 holds for a fixed $\gamma > 0$, then no nonzero Type I blow-up limit can have a centered representative satisfying the controlled-swirl condition with exponent γ .*

Proof. This is the hypothesis combined with the nontriviality Theorem 2.3. $\square$

5.2. Weighted velocity decay.

Definition 5.4 (Weighted velocity-decay condition). Fix $m > 0$. A centered ancient solution V satisfies the weighted velocity-decay condition with exponent m if

$$\sup_{\tau \in \mathbb{R}} \int_{\mathbb{R}^3} (1 + |y|^2)^m |V(y, \tau)|^2 dy < \infty.$$

Hypothesis 5.5 (Fixed-exponent weighted velocity Liouville theorem). For the chosen exponent $m > 0$, every Type I ancient solution whose centered representative satisfies the weighted velocity-decay condition with exponent m is identically zero.

Proposition 5.6 (Conditional weighted velocity vanishing). *If Theorem 5.5 holds for a fixed $m > 0$, then no nonzero Type I blow-up limit can have a centered representative satisfying the weighted velocity-decay condition with exponent m .*

Proof. This is Theorem 5.5 plus Theorem 2.3. $\square$

Remark 5.7 (Weighted velocity versus weighted vorticity). The weighted velocity condition is not covered by the Gallay–Wayne theorem below. Gallay–Wayne applies only to ancient trajectories whose rescaled vorticity remains in a sufficiently small ball of $L_\sigma^2(m)$. Thus Theorem 5.5 remains an independent fixed-exponent Liouville hypothesis unless weighted velocity decay is shown to imply the corresponding perturbative weighted-vorticity condition.

6. PERTURBATIVE WEIGHTED-VORTICITY LYAPUNOV RIGIDITY

6.1. The rescaled vorticity equation. In this section τ is the forward self-similar time used in the Gallay–Wayne vorticity formulation. The ancient trajectories are indexed by all $\tau \in \mathbb{R}$, and the forward semiflow acts by increasing τ .

Let $w = \nabla \times v$. The rescaled vorticity equation is

$$(6.1) \quad \partial_\tau w = \Lambda w - (v \cdot \nabla)w + (w \cdot \nabla)v, \quad \nabla \cdot w = 0,$$

where

$$(6.2) \quad \Lambda w = \Delta w + \frac{1}{2} \xi \cdot \nabla w + w,$$

and v is recovered from w by the three-dimensional Biot–Savart law. For $m \geq 0$, set

$$L_\sigma^2(m) = \{f \in L^2(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot f = 0, \quad \|f\|_m < \infty\},$$

where

$$\|f\|_m^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2m} |f(\xi)|^2 d\xi.$$

6.2. Galloway–Wayne small-data decay.

Theorem 6.1 (Galloway–Wayne small-data decay). *Let $0 < \mu \leq 1$ and $m > 2\mu + \frac{1}{2}$. There exist constants $\rho_m > 0$ and $C_m \geq 1$ such that if $\|w_0\|_m \leq \rho_m$, then (6.1) has a unique global mild solution $w(\tau) = \Phi_\tau w_0$ in $C([0, \infty); L_\sigma^2(m))$ and*

$$\|\Phi_\tau w_0\|_m \leq C_m e^{-\mu\tau} \|w_0\|_m, \quad \tau \geq 0.$$

In particular, if $m > \frac{7}{2}$, one may use $\mu = 1$.

Proof. This is the Galloway–Wayne weighted small-data theorem for the rescaled three-dimensional vorticity equation [6]. The only conclusions used below are existence, uniqueness, the semiflow property, and the displayed exponential decay. $\square$

6.3. The Lyapunov functional.

Definition 6.2 (Perturbative weighted-vorticity condition). Fix $m > \frac{7}{2}$, and let ρ_m be the radius in Theorem 6.1. An ancient vorticity trajectory satisfies the perturbative weighted-vorticity condition with exponent m if it is a function

$$w \in C(\mathbb{R}; L_\sigma^2(m))$$

that solves (6.1) for all $\tau \in \mathbb{R}$ and satisfies

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_m \leq \rho_m.$$

Solving for all τ means that every time shift is compatible with the Galloway–Wayne forward mild semiflow.

Proposition 6.3 (Semiflow Lyapunov functional). *Fix $m > \frac{7}{2}$. For $\|w_0\|_m \leq \rho_m$, define*

$$(6.3) \quad \mathcal{L}_m(w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2.$$

Then $\mathcal{L}_m$ is well-defined and satisfies

$$(6.4) \quad \|w_0\|_m^2 \leq \mathcal{L}_m(w_0) \leq C_m^2 \|w_0\|_m^2.$$

Moreover, for every $s \geq 0$,

$$(6.5) \quad \mathcal{L}_m(\Phi_s w_0) \leq e^{-2s} \mathcal{L}_m(w_0).$$

Proof. The Galloway–Wayne estimate with $\mu = 1$ gives

$$e^{2\tau} \|\Phi_\tau w_0\|_m^2 \leq C_m^2 \|w_0\|_m^2,$$

which proves finiteness and the upper bound. The lower bound is the value at $\tau = 0$. By uniqueness,

$$\Phi_\tau(\Phi_s w_0) = \Phi_{\tau+s} w_0.$$

Hence

$$\mathcal{L}_m(\Phi_s w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_{\tau+s} w_0\|_m^2 = e^{-2s} \sup_{\sigma \geq s} e^{2\sigma} \|\Phi_\sigma w_0\|_m^2 \leq e^{-2s} \mathcal{L}_m(w_0).$$

□

6.4. Ancient rigidity in the small weighted ball.

Proof of Theorem 1.2. Let w be an ancient solution in the small ball and fix $\tau_0 \in \mathbb{R}$. For $s < \tau_0$, the shifted function

$$W_s(\theta) = w(s + \theta), \quad \theta \geq 0,$$

is the Galloway–Wayne mild solution with initial datum $w(s)$. The small-ball hypothesis gives $\|w(s)\|_m \leq \rho_m$, so the forward solution is in the domain of Φ_θ , and uniqueness gives

$$W_s(\theta) = \Phi_\theta w(s), \quad \theta \geq 0.$$

Choosing $\theta = \tau_0 - s$ gives

$$w(\tau_0) = \Phi_{\tau_0-s} w(s).$$

Since $\|w(\tau_0)\|_m \leq \rho_m$, the Lyapunov functional is also defined at $w(\tau_0)$. Applying Theorem 6.3 with forward time $\tau_0 - s$,

$$\mathcal{L}_m(w(\tau_0)) \leq e^{-2(\tau_0-s)} \mathcal{L}_m(w(s)) \leq C_m^2 \rho_m^2 e^{-2(\tau_0-s)}.$$

Letting $s \rightarrow -\infty$ gives $\mathcal{L}_m(w(\tau_0)) = 0$. The coercive lower bound (6.4) gives $w(\tau_0) = 0$. Since τ_0 was arbitrary, $w \equiv 0$. □

Remark 6.4 (Relation to weighted velocity decay). This theorem applies only to Type I blow-up limits whose vorticity satisfies the Galloway–Wayne perturbative weighted condition. It does not exclude all weighted velocity-decay conditions.

7. THE BURGERS-VORTEX EQUATION

This section proves a rigidity theorem for the strained Burgers-vortex equation. This result does not by itself apply to the unforced Navier–Stokes equations. The final corollary states the additional association assumption needed to transfer the conclusion to an unforced ancient-solution class.

7.1. The strained vorticity equation. Let $x = (x_h, x_3) \in \mathbb{R}^2 \times \mathbb{R}$ and

$$A = \text{diag} \left(-\frac{1}{2}, -\frac{1}{2}, 1 \right).$$

The Burgers-vortex vorticity equation is

$$(7.1) \quad \partial_\tau \Omega + (U \cdot \nabla) \Omega - (\Omega \cdot \nabla) U = \Delta \Omega - (Ax \cdot \nabla) \Omega + A \Omega, \quad \nabla \cdot \Omega = 0,$$

with U recovered from Ω by the Biot–Savart law. It has the stationary family

$$(7.2) \quad \Omega = \alpha G, \quad G(x) = \begin{pmatrix} 0 \\ 0 \\ g(x_h) \end{pmatrix}, \quad g(x_h) = \frac{1}{4\pi} e^{-|x_h|^2/4}, \quad \alpha \in \mathbb{R}.$$

7.2. Weighted horizontal spaces. For $m > 1$, let $L_h^2(m)$ denote the horizontal weighted space with norm

$$\|f\|_{L_h^2(m)}^2 = \int_{\mathbb{R}^2} |f(x_h)|^2 \left(1 + \frac{|x_h|^2}{4m}\right)^m dx_h.$$

Set

$$X(m) = BC(\mathbb{R}_{x_3}; L_h^2(m)), \quad X_0(m) = \left\{ f \in X(m) : \int_{\mathbb{R}^2} f(x_h, x_3) dx_h = 0 \text{ for every } x_3 \right\},$$

and

$$\mathcal{X}_{\text{full}}(m) = X(m)^3, \quad \mathcal{X}_{\text{mod}}(m) = X(m) \times X(m) \times X_0(m).$$

7.3. Admissible Burgers-vortex solutions.

Definition 7.1 (Admissible Burgers-vortex solutions). A mild solution $\Omega : \mathbb{R} \rightarrow \mathcal{X}_{\text{full}}(m)$ is called admissible if the following operations are justified on every compact time interval:

- (i) horizontal integrations by parts against constants and against x_h are valid, with no boundary contribution at $|x_h| = \infty$;
- (ii) the identity

$$\int_{\mathbb{R}^2} [(U \cdot \nabla) \Omega - (\Omega \cdot \nabla) U]_3 dx_h = 0$$

holds in distributions in x_3 and τ ;

- (iii) the map

$$\tau \mapsto \int_{\mathbb{R}^2} \Omega_3(x_h, x_3, \tau) dx_h$$

is absolutely continuous after testing in x_3 .

Smooth rapidly decaying solutions are admissible.

7.4. Circulation coordinate and conservation.

Lemma 7.2 (Circulation coordinate). *Let $m > 1$ and let $\Omega \in \mathcal{X}_{\text{full}}(m)$ be divergence-free. Assume that the horizontal integration by parts*

$$\int_{\mathbb{R}^2} \nabla_h \cdot \Omega_h(x_h, x_3) dx_h = 0$$

is valid. Then

$$\alpha[\Omega](x_3) = \int_{\mathbb{R}^2} \Omega_3(x_h, x_3) dx_h$$

is well-defined and independent of x_3 . Writing the common value as $\alpha[\Omega]$, one has

$$Q\Omega := \Omega - \alpha[\Omega]G \in \mathcal{X}_{\text{mod}}(m).$$

Proof. Since $m > 1$, $L_h^2(m) \hookrightarrow L^1(\mathbb{R}^2)$ by Cauchy-Schwarz:

$$\int_{\mathbb{R}^2} |f(x_h)| dx_h \leq \|f\|_{L_h^2(m)} \left(\int_{\mathbb{R}^2} \left(1 + \frac{|x_h|^2}{4m}\right)^{-m} dx_h \right)^{1/2},$$

and the last integral is finite in two dimensions exactly because $m > 1$. Thus $\alpha[\Omega](x_3)$ is well-defined. Using $\nabla \cdot \Omega = 0$,

$$\partial_{x_3} \alpha[\Omega](x_3) = \int_{\mathbb{R}^2} \partial_{x_3} \Omega_3 dx_h = - \int_{\mathbb{R}^2} \nabla_h \cdot \Omega_h dx_h = 0.$$

Also $\int_{\mathbb{R}^2} g(x_h) dx_h = 1$, so subtracting $\alpha[\Omega]G$ removes the horizontal mean of the third component. $\square$

Lemma 7.3 (Conservation of circulation). *Let $m > 1$, and let $\Omega(\tau)$ be an admissible mild solution of (7.1) with values in $\mathcal{X}_{\text{full}}(m)$. Then $\alpha[\Omega(\tau)]$ is independent of τ . If $\bar{\alpha}$ denotes this constant, then*

$$\omega(\tau) = \Omega(\tau) - \bar{\alpha}G$$

belongs to $\mathcal{X}_{\text{mod}}(m)$ for every τ and solves the fixed-circulation perturbation equation around $\bar{\alpha}G$.

Proof. Integrate the third component of (7.1) over x_h . The nonlinear term is a horizontal divergence:

$$(U \cdot \nabla)\Omega - (\Omega \cdot \nabla)U = -\nabla \times (U \times \Omega),$$

and the third component of a curl is a horizontal divergence. It therefore has zero horizontal integral by admissibility. For the linear part,

$$[\Delta\Omega - (Ax \cdot \nabla)\Omega + A\Omega]_3 = \Delta_h \Omega_3 + \partial_{x_3}^2 \Omega_3 + \frac{1}{2}x_h \cdot \nabla_h \Omega_3 - x_3 \partial_{x_3} \Omega_3 + \Omega_3.$$

The Δ_h term integrates to zero, and

$$\int_{\mathbb{R}^2} \frac{1}{2}x_h \cdot \nabla_h \Omega_3 dx_h = - \int_{\mathbb{R}^2} \Omega_3 dx_h,$$

which cancels the $+\Omega_3$ term. Hence

$$\frac{d}{d\tau} \alpha[\Omega(\tau)] = \partial_{x_3}^2 \alpha[\Omega(\tau)] - x_3 \partial_{x_3} \alpha[\Omega(\tau)].$$

By Theorem 7.2, the right-hand side is zero. $\square$

7.5. Galloway–Maekawa stability.

Theorem 7.4 (Galloway–Maekawa stability). *Let $m > 2$ and $\alpha \in \mathbb{R}$. There exist constants $\delta_{\alpha,m} > 0$ and $C_{\alpha,m} \geq 1$ such that every divergence-free perturbation*

$$\omega_0 \in \mathcal{X}_{\text{mod}}(m), \quad \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\alpha,m},$$

generates a unique global mild solution of the fixed-circulation perturbation equation around αG , denoted $\Phi_\tau^\alpha \omega_0$, and

$$\|\Phi_\tau^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \leq C_{\alpha,m} e^{-\tau/2} \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \quad (\tau \geq 0).$$

Proof. This is the Galloway–Maekawa nonlinear stability theorem for Burgers vortices in horizontal weighted spaces [8]. We use only existence, uniqueness, the fixed-circulation semiflow property, and the displayed exponential decay. $\square$

7.6. Lyapunov functional near a Burgers vortex.

Proposition 7.5 (Lyapunov functional relative to a Burgers vortex). *Fix $m > 2$ and $\alpha \in \mathbb{R}$. On the ball*

$$\|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\alpha,m}$$

define

$$(7.3) \quad \mathcal{B}_{\alpha,m}(\omega_0) = \sup_{\tau \geq 0} e^\tau \|\Phi_\tau^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2.$$

Then

$$\|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2 \leq \mathcal{B}_{\alpha,m}(\omega_0) \leq C_{\alpha,m}^2 \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2,$$

and

$$\mathcal{B}_{\alpha,m}(\Phi_s^\alpha \omega_0) \leq e^{-s} \mathcal{B}_{\alpha,m}(\omega_0) \quad (s \geq 0).$$

Proof. Gally–Maekawa stability gives, for every $\tau \geq 0$,

$$e^\tau \|\Phi_\tau^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2 \leq C_{\alpha,m}^2 \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2.$$

Taking the supremum over $\tau \geq 0$ proves that $\mathcal{B}_{\alpha,m}$ is finite and gives the upper coercive bound. The lower bound follows from the value at $\tau = 0$, because $\Phi_0^\alpha \omega_0 = \omega_0$.

For $s \geq 0$, uniqueness of the fixed-circulation perturbation equation gives the semiflow identity

$$\Phi_\tau^\alpha(\Phi_s^\alpha \omega_0) = \Phi_{\tau+s}^\alpha \omega_0.$$

Therefore

$$\mathcal{B}_{\alpha,m}(\Phi_s^\alpha \omega_0) = \sup_{\tau \geq 0} e^\tau \|\Phi_{\tau+s}^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2.$$

With $\sigma = \tau + s$, this is

$$e^{-s} \sup_{\sigma \geq s} e^\sigma \|\Phi_\sigma^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2 \leq e^{-s} \mathcal{B}_{\alpha,m}(\omega_0).$$

□

7.7. Ancient rigidity near the Burgers-vortex family.

Definition 7.6 (Small ancient solutions near Burgers vortices). Fix $m > 2$. We say that an admissible ancient mild solution $\Omega : \mathbb{R} \rightarrow \mathcal{X}_{\text{full}}(m)$ of (7.1) stays in the Gally–Maekawa neighborhood of the Burgers-vortex family if, with $\bar{\alpha} = \alpha[\Omega(\tau)]$,

$$\sup_{\tau \in \mathbb{R}} \|\Omega(\tau) - \bar{\alpha}G\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\bar{\alpha},m}.$$

The term “ancient mild” includes compatibility with the Gally–Maekawa forward mild semiflow after every time shift.

Proof of Theorem 1.3. By Theorem 7.3, $\bar{\alpha}$ is independent of time and

$$\omega(\tau) = \Omega(\tau) - \bar{\alpha}G$$

is an ancient solution of the fixed-circulation perturbation equation in the small ball. Fix $\tau_0 \in \mathbb{R}$ and $s < \tau_0$. By the ancient mild compatibility, the shifted perturbation

$$\omega_s(\theta) = \omega(s + \theta), \quad \theta \geq 0,$$

is the Gally–Maekawa mild solution launched from $\omega(s)$. The neighborhood hypothesis gives

$$\|\omega(s)\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\bar{\alpha},m},$$

so uniqueness of the fixed-circulation semiflow gives

$$\omega(\tau_0) = \Phi_{\tau_0-s}^{\bar{\alpha}} \omega(s).$$

Using Theorem 7.5,

$$\mathcal{B}_{\bar{\alpha},m}(\omega(\tau_0)) \leq e^{-(\tau_0-s)} \mathcal{B}_{\bar{\alpha},m}(\omega(s)) \leq e^{-(\tau_0-s)} C_{\bar{\alpha},m}^2 \delta_{\bar{\alpha},m}^2.$$

Letting $s \rightarrow -\infty$ gives $\mathcal{B}_{\bar{\alpha},m}(\omega(\tau_0)) = 0$. The coercive lower bound gives $\omega(\tau_0) = 0$. Since τ_0 was arbitrary, $\Omega(\tau) \equiv \bar{\alpha}G$. □

7.8. Conditional association with unforced Navier–Stokes.

Assumption 7.7 (Association with the Burgers-vortex equation). Let $\mathcal{C}$ be a class of ancient solutions of an unforced or renormalized Navier–Stokes problem. Assume that $m > 2$ and that each $u \in \mathcal{C}$ is equipped with a field

$$\Omega_u \in C_{\text{loc}}(\mathbb{R}; \mathcal{X}_{\text{full}}(m))$$

and a nonnegative function $\varepsilon_u(\tau)$ such that:

- (i) Ω_u is an admissible ancient mild solution of (7.1) in the sense of Theorem 7.1;
- (ii) with $\bar{\alpha} = \alpha[\Omega_u]$,

$$\|\Omega_u(\tau) - \bar{\alpha}G\|_{\mathcal{X}_{\text{mod}}(m)} \leq C_* \varepsilon_u(\tau) \quad (\tau \in \mathbb{R});$$

- (iii) $\sup_{\tau \in \mathbb{R}} \varepsilon_u(\tau) \leq \delta_{\bar{\alpha}, m}/C_*$.

Corollary 7.8 (Conditional Burgers-vortex conclusion). *Under Theorem 7.7, the associated field for every $u \in \mathcal{C}$ is an exact Burgers vortex:*

$$\Omega_u(\tau) \equiv \bar{\alpha}G.$$

Proof. The assumptions place Ω_u in the small Galloway–Maekawa neighborhood defined in Theorem 7.6. Applying Theorem 1.3 gives the conclusion. $\square$

8. THE SCALAR Γ IDENTITY AND CUTOFF CRITERION

8.1. The equation for Γ .

Lemma 8.1 (Swirl equation). *Let u be a smooth axisymmetric solution of (1.1). Then $\Gamma = ru_\theta$ satisfies*

$$(8.1) \quad \partial_t \Gamma + b \cdot \nabla \Gamma + \frac{2}{r} \partial_r \Gamma = \Delta \Gamma, \quad b = u_r e_r + u_z e_z.$$

Proof. The angular component of (1.1) is

$$\partial_t u_\theta + u_r \partial_r u_\theta + u_z \partial_z u_\theta + \frac{u_r}{r} u_\theta = \left(\partial_r^2 + \frac{1}{r} \partial_r + \partial_z^2 - \frac{1}{r^2} \right) u_\theta.$$

Multiplying by r , the transport terms become

$$ru_r \partial_r u_\theta + u_r u_\theta = u_r \partial_r (ru_\theta) = u_r \partial_r \Gamma, \quad ru_z \partial_z u_\theta = u_z \partial_z \Gamma.$$

For the radial diffusion, use $u_\theta = \Gamma/r$. Then

$$\partial_r u_\theta = \frac{1}{r} \partial_r \Gamma - \frac{1}{r^2} \Gamma, \quad \partial_r^2 u_\theta = \frac{1}{r} \partial_r^2 \Gamma - \frac{2}{r^2} \partial_r \Gamma + \frac{2}{r^3} \Gamma.$$

Consequently

$$r \left(\partial_r^2 + \frac{1}{r} \partial_r - \frac{1}{r^2} \right) u_\theta = \partial_r^2 \Gamma - \frac{1}{r} \partial_r \Gamma.$$

The axial diffusion satisfies $r \partial_z^2 u_\theta = \partial_z^2 \Gamma$. Since

$$\Delta \Gamma = \partial_r^2 \Gamma + \frac{1}{r} \partial_r \Gamma + \partial_z^2 \Gamma,$$

the total diffusion term after multiplication by r is

$$\partial_r^2 \Gamma - \frac{1}{r} \partial_r \Gamma + \partial_z^2 \Gamma = \Delta \Gamma - \frac{2}{r} \partial_r \Gamma.$$

Substitution gives

$$\partial_t \Gamma + b \cdot \nabla \Gamma = \Delta \Gamma - \frac{2}{r} \partial_r \Gamma,$$

which is (8.1). □

8.2. The weighted scalar identity.

Proposition 8.2 (Weighted scalar identity). *Let u be a smooth axisymmetric solution of (1.1). Set $W = \Gamma$, and let $\phi = \phi(r, z, t) \geq 0$ be a smooth compactly supported axisymmetric weight. Then*

$$(8.2) \quad \frac{1}{2} \frac{d}{dt} \int_{\mathbb{R}^3} W^2 \phi \, dx + \int_{\mathbb{R}^3} |\nabla W|^2 \phi \, dx = \frac{1}{2} \int_{\mathbb{R}^3} W^2 \left(\partial_t \phi + \Delta \phi + b \cdot \nabla \phi + \frac{2}{r} \partial_r \phi \right) dx.$$

If $W = \Gamma - c$, the same identity holds provided $c = 0$ or ϕ vanishes near the axis. Without this restriction, the right-hand side must be augmented by the axis boundary term

$$2\pi c^2 \int_{\mathbb{R}} \phi(0, z, t) \, dz.$$

Proof. From Theorem 8.1,

$$\partial_t W = \Delta W - b \cdot \nabla W - \frac{2}{r} \partial_r W.$$

Multiply by $W\phi$ and integrate. The time derivative gives

$$\int W \partial_t W \phi \, dx = \frac{1}{2} \frac{d}{dt} \int W^2 \phi \, dx - \frac{1}{2} \int W^2 \partial_t \phi \, dx.$$

The Laplacian gives

$$\int W \Delta W \phi \, dx = - \int |\nabla W|^2 \phi \, dx + \frac{1}{2} \int W^2 \Delta \phi \, dx.$$

Because u is divergence-free and b is its meridional part, $\nabla \cdot b = 0$ in the three-dimensional sense:

$$\nabla \cdot b = \frac{1}{r} \partial_r(r u_r) + \partial_z u_z = \nabla \cdot u = 0.$$

Thus, using compact support of ϕ ,

$$- \int W(b \cdot \nabla W) \phi \, dx = \frac{1}{2} \int W^2 b \cdot \nabla \phi \, dx.$$

Finally, in cylindrical variables $dx = r \, dr \, d\theta \, dz$,

$$- \int W \frac{2}{r} \partial_r W \phi \, dx = - \int_0^{2\pi} \int_{\mathbb{R}} \int_0^\infty \partial_r(W^2) \phi \, dr \, dz \, d\theta.$$

For $W = \Gamma$, smoothness gives $\Gamma = O(r^2)$ at the axis, so the boundary term at $r = 0$ vanishes. More explicitly, after integrating on $\delta < r < \infty$ and letting $\delta \downarrow 0$, the boundary contribution is

$$\int_0^{2\pi} \int_{\mathbb{R}} W(\delta, z, t)^2 \phi(\delta, z, t) \, dz \, d\theta \longrightarrow 0.$$

Therefore the last display equals

$$\int W^2 \frac{1}{r} \partial_r \phi \, dx = \frac{1}{2} \int W^2 \frac{2}{r} \partial_r \phi \, dx.$$

Combining the terms proves (8.2). If $W = \Gamma - c$, then $W(0, z, t) = -c$. The same integration on $\delta < r < \infty$ leaves the limit

$$\int_0^{2\pi} \int_{\mathbb{R}} c^2 \phi(0, z, t) \, dz \, d\theta = 2\pi c^2 \int_{\mathbb{R}} \phi(0, z, t) \, dz,$$

which is absent exactly when $c = 0$ or ϕ avoids the axis. □

8.3. Anisotropic product cutoffs.

$$\phi_{R,Z,\zeta}(r, z, t) = \lambda_R(r) \eta_Z(z - \zeta(t)),$$

where λ_R, η_Z are smooth cutoffs and ζ is a smooth axial center.

Proposition 8.3 (Cutoff-error decomposition). *Let $W = \Gamma$. With the notation above,*

$$(8.3) \quad \frac{1}{2} \frac{d}{dt} \int W^2 \lambda_R \eta_Z dx + \int |\nabla W|^2 \lambda_R \eta_Z dx = \frac{1}{2} \int W^2 \mathcal{E}_{R,Z,\zeta}[u] dx,$$

where

$$(8.4) \quad \mathcal{E}_{R,Z,\zeta}[u] = \eta_Z \left(\lambda_R'' + \frac{3}{r} \lambda_R' + u_r \lambda_R' \right) + \lambda_R \left(\eta_Z'' + (u_z - \dot{\zeta}) \eta_Z' \right).$$

Proof. For an axisymmetric scalar,

$$\Delta(\lambda_R \eta_Z) = \left(\lambda_R'' + \frac{1}{r} \lambda_R' \right) \eta_Z + \lambda_R \eta_Z''.$$

Also

$$\frac{2}{r} \partial_r (\lambda_R \eta_Z) = \frac{2}{r} \lambda_R' \eta_Z, \quad b \cdot \nabla (\lambda_R \eta_Z) = u_r \lambda_R' \eta_Z + u_z \lambda_R \eta_Z',$$

and

$$\partial_t (\lambda_R \eta_Z) = -\dot{\zeta} \lambda_R \eta_Z'.$$

Adding these four contributions gives (8.4); substitution into [Theorem 8.2](#) gives (8.3). $\square$

Remark 8.4 (The radial geometry). The differential expression

$$\lambda_R'' + \frac{3}{r} \lambda_R'$$

is the radial Laplacian in four dimensions. Thus the natural radial barrier for the Γ equation is adapted to $\partial_r^2 + 3r^{-1} \partial_r$. The cutoff errors in (8.4) are precisely radial transport, axial drift mismatch, and the two cutoff-curvature terms.

8.4. Coercive expanding-cutoff control.

Assumption 8.5 (Coercive expanding-cutoff control). Let u be a bounded ancient mild axisymmetric solution with bounded Γ . Assume in addition that u is smooth and that the identities [Theorems 8.1](#) to [8.3](#) hold for the weights below. There exist expanding smooth weights

$$\phi_n(r, z, t) = \lambda_n(r) \eta_n(z - \zeta_n(t)), \quad 0 \leq \phi_n \leq 1, \quad \phi_n \uparrow 1$$

locally uniformly on compact subsets of $\mathbb{R}^3 \times (-\infty, 0)$, compactly supported in space for each fixed time. There are constants $a_n > 0$, $M_n < \infty$, and $\varepsilon_n \geq 0$, with $\varepsilon_n/a_n \rightarrow 0$, such that

$$L_n(t) := \int_{\mathbb{R}^3} \Gamma^2 \phi_n(t) dx < \infty$$

for every $t < 0$, L_n is locally absolutely continuous, and, for each n , the following estimates hold for a.e. $\sigma < 0$:

$$(8.5) \quad \int_{\mathbb{R}^3} |\nabla \Gamma|^2 \phi_n(\sigma) dx \geq a_n L_n(\sigma),$$

$$(8.6) \quad \left| \int_{\mathbb{R}^3} \Gamma^2 \mathcal{E}_n[u] dx \right| \leq 2\varepsilon_n,$$

and

$$(8.7) \quad \sup_{\sigma < t} L_n(\sigma) \leq M_n.$$

Here $\mathcal{E}_n[u]$ is the cutoff-error density associated with ϕ_n .

8.5. Scalar conclusion from expanding cutoffs.

Proof of Theorem 1.4. Fix n . Integrating (8.3) from s to t , with $-\infty < s < t < 0$, gives

$$\frac{1}{2}L_n(t) - \frac{1}{2}L_n(s) + \int_s^t \int_{\mathbb{R}^3} |\nabla \Gamma|^2 \phi_n \, dx \, d\sigma = \frac{1}{2} \int_s^t \int_{\mathbb{R}^3} \Gamma^2 \mathcal{E}_n[u] \, dx \, d\sigma.$$

Using (8.5) and (8.6), this implies

$$\frac{1}{2}L_n(t) - \frac{1}{2}L_n(s) + a_n \int_s^t L_n(\sigma) \, d\sigma \leq \varepsilon_n(t-s).$$

Since L_n is locally absolutely continuous, the preceding integral inequality yields the differential inequality

$$\frac{d}{dt}L_n(t) + 2a_n L_n(t) \leq 2\varepsilon_n$$

for a.e. $t < 0$. Multiplying by $e^{2a_n t}$ and integrating from s to t gives

$$L_n(t) \leq e^{-2a_n(t-s)} L_n(s) + \frac{\varepsilon_n}{a_n} \left(1 - e^{-2a_n(t-s)}\right).$$

Using the past bound (8.7) and letting $s \rightarrow -\infty$ yields

$$L_n(t) \leq \frac{\varepsilon_n}{a_n}.$$

Let $K \subset \mathbb{R}^3$ be compact and fix $t < 0$. Since $\phi_n \uparrow 1$ locally uniformly in space-time, for every $\eta \in (0, 1)$ there is $n_0 = n_0(K, t, \eta)$ such that

$$\phi_n(x, t) \geq 1 - \eta \quad (x \in K, n \geq n_0).$$

Therefore, for $n \geq n_0$,

$$(1 - \eta) \int_K \Gamma(x, t)^2 \, dx \leq L_n(t) \leq \frac{\varepsilon_n}{a_n}.$$

Letting $n \rightarrow \infty$ and using $\varepsilon_n/a_n \rightarrow 0$ gives

$$\int_K \Gamma(x, t)^2 \, dx = 0$$

for every compact K and every $t < 0$. Since Γ is continuous by the smoothness in Theorem 8.5, local L^2 -vanishing on every compact set implies $\Gamma(x, t) = 0$ pointwise for every x and $t < 0$. Thus $\Gamma \equiv 0$. $\square$

8.6. Consequence of scalar vanishing.

Corollary 8.6 (Consequence of scalar vanishing). *Let u be as in Theorem 1.4. If in addition u is a nonzero Type I blow-up limit in the same physical frame, then $\Gamma \equiv 0$, $u_\theta \equiv 0$, and $u \equiv 0$, contradicting the nontriviality of the blow-up limit.*

Proof. Theorem 1.4 gives $\Gamma \equiv 0$. Then $u_\theta = 0$ for $r > 0$, so the no-swirl Type I vanishing result Theorem 4.1 applies. $\square$

9. SUMMARY OF PDE CONSEQUENCES

9.1. Zero conclusions.

Theorem 9.1 (Zero conclusions for Type I blow-up limits). *The following hypotheses cannot hold for a nonzero Type I blow-up limit:*

- (i) *axisymmetry and $U_\theta \equiv 0$;*
- (ii) *axisymmetry and the pointwise bound $|U(x, t)| \leq C/r$;*
- (iii) *axisymmetry and finite- L^p control of the swirl scalar,*

$$\Gamma \in L_t^\infty L_x^p, \quad 1 \leq p < \infty;$$

- (iv) *axisymmetry, z -periodicity, and bounded Γ ;*
- (v) *the Galloway–Wayne small weighted-vorticity condition, whenever the velocity is recovered from the corresponding vorticity by the Biot–Savart law;*
- (vi) *bounded Γ together with the expanding-cutoff hypotheses in [Theorem 8.5](#).*

Proof. Items (i)–(iv) are [Theorem 1.1](#). Item (v) is [Theorem 1.2](#); since the velocity in that case is recovered from w by the Biot–Savart law, $w \equiv 0$ gives the zero velocity. Item (vi) is [Theorem 8.6](#). In each Type I case, the zero conclusion contradicts [Theorem 2.3](#). $\square$

9.2. Burgers-vortex rigidity.

Theorem 9.2 (Burgers-vortex rigidity). *Every admissible ancient solution of the strained Burgers-vortex equation that stays in the Galloway–Maekawa neighborhood of the Burgers-vortex family is a stationary Burgers vortex αG . Under the association assumption [Theorem 7.7](#), the same conclusion holds for the associated field Ω_u .*

Proof. The rigidity statement is [Theorem 1.3](#); the conditional association statement is [Theorem 7.8](#). $\square$

9.3. Conditional Liouville hypotheses.

Proposition 9.3 (Conditional Liouville hypotheses). *Under [Theorem 5.2](#), no nonzero Type I blow-up limit can satisfy the controlled-swirl condition with exponent γ . Under [Theorem 5.5](#), no nonzero Type I blow-up limit can satisfy the weighted velocity-decay condition with exponent m .*

Proof. This is [Theorems 5.3](#) and [5.6](#). $\square$

9.4. Bounded swirl without cutoff control.

Remark 9.4 (Nonperiodic bounded swirl without the cutoff criterion). The full class of bounded ancient axisymmetric unforced Navier–Stokes solutions with nontrivial bounded Γ , no finite- L^p decay of Γ , no axial periodicity, and no verified expanding-cutoff estimate is not treated unconditionally here. The identity (8.3) reduces this case to explicit transport and cutoff terms, but it does not eliminate those terms without the additional estimates in [Theorem 8.5](#).

9.5. Consequences for Type I blow-up arguments.

Proof of [Theorem 1.5](#). The axisymmetric cases are ruled out by [Theorem 1.1](#). The perturbative weighted-vorticity case is ruled out by [Theorem 1.2](#) once the corresponding rescaled vorticity satisfies the small weighted condition; the Biot–Savart reconstruction then gives zero velocity. The bounded-swirl case under [Theorem 8.5](#) is ruled out by [Theorem 8.6](#). These zero conclusions contradict the nontriviality of a Type I blow-up limit. The Burgers-vortex conclusion is separate: under [Theorem 7.7](#), [Theorem 7.8](#) gives $\Omega_U(\tau) \equiv \bar{\alpha}G$. This strained-equation theorem does not, by itself, assert a zero-solution conclusion for the unforced equations. $\square$

Theorem 9.5 (Combined zero conclusion). *Let U be a Type I ancient solution obtained as a nonzero Type I blow-up limit. If U satisfies one of the hypotheses in [Theorem 9.1](#), or one of the conditional hypotheses in [Theorem 9.3](#) for which the stated Liouville theorem is available, then $U \equiv 0$, a contradiction. The Burgers-vortex equation contributes only the separate rigidity statement of [Theorem 9.2](#).*

Proof. This is [Theorem 9.1](#) and [Theorem 9.3](#), with the Burgers-vortex assertion recorded separately in [Theorem 9.2](#). $\square$

9.6. Summary table.

Hypothesis	Conclusion	Analytic ingredient
no swirl	$U \equiv 0$	KNSS plus Type I constant removal
pointwise $1/r$	$U \equiv 0$	KNSS
$\Gamma \in L_t^\infty L_x^p$	$U \equiv 0$	LZZ plus Type I constant removal
periodic bounded Γ	$U \equiv 0$	LRZ plus Type I constant removal
controlled swirl $\rho^\gamma V_\theta$	conditional	Theorem 5.2
weighted velocity decay	conditional	Theorem 5.5
small weighted vorticity	$w \equiv 0$	Gallay–Wayne Lyapunov functional
near Burgers vortices	$\Omega = \alpha G$	Gallay–Maekawa Lyapunov functional
association with Burgers vortices	conditional	Theorem 7.7
bounded nonperiodic $\Gamma \in L^\infty$	conditional	Theorem 8.5 needed
bounded Γ plus cutoff control	$\Gamma \equiv 0$	scalar identity plus no-swirl theorem

The no-swirl, pointwise scale-invariant, finite- L^p -swirl, and periodic bounded-swirl cases are covered by known axisymmetric Liouville theorems. The perturbative weighted-vorticity case is covered by a Lyapunov functional derived from Gallay–Wayne decay, and small ancient perturbations of Burgers vortices are rigid for the strained equation by Gallay–Maekawa stability. The remaining nonperiodic bounded-swirl case is reduced here to the coercive expanding-cutoff criterion for the scalar $\Gamma = ru_\theta$. Thus every zero conclusion used in the Type I application rests on an explicit PDE hypothesis stated in this paper.

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